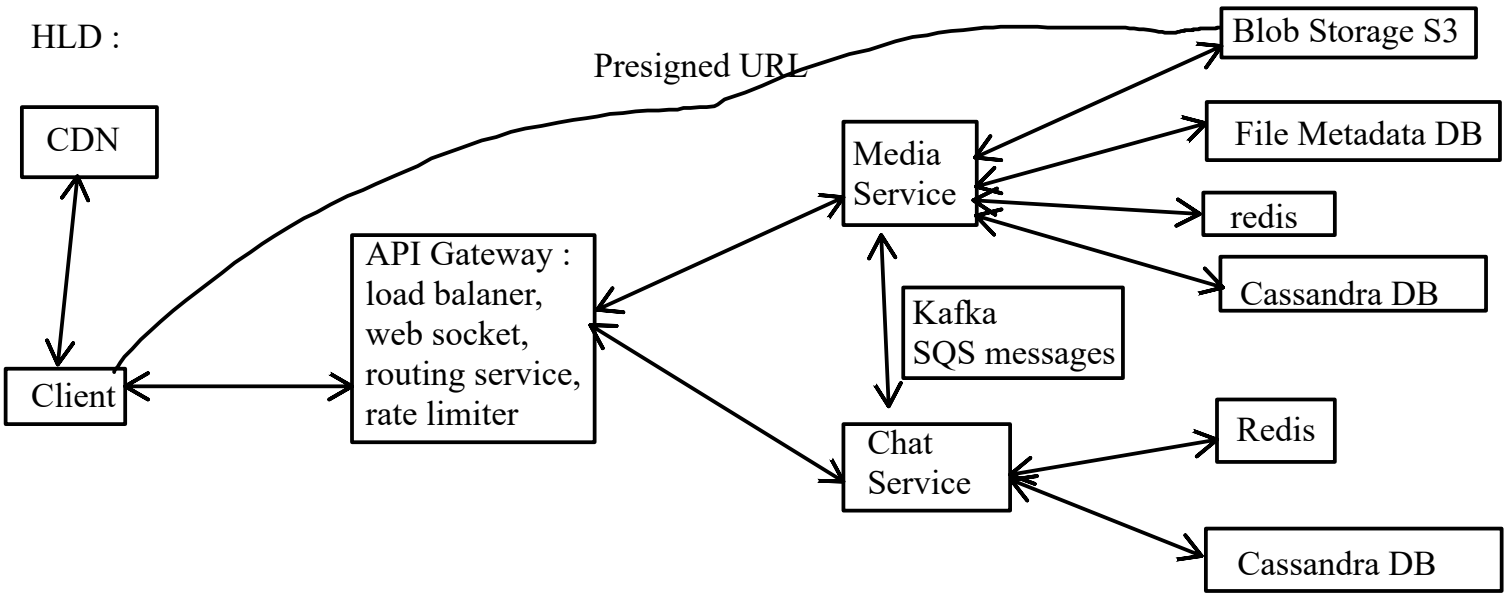
Whats App

Functional requirements:
user should be able to send message & media based on his preference,
user should be able to search for chats,
user should be able to make calls

Non-functional requirements:
Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency,
high throughput,
CAP theorem (here eventual consistency is fine , but here we need 99.99% uptime
i.e Availability>>Consistency)

Core Entities:
user,
message,
media,
chat

API :
POST:/post/message
{
userID,
{filemetadata}
}
GET:/search/{chatID}

HLD :